GHraisoni COLLEGE
Engineering and Management
Nagpur

RSOC-2026

RSOC 2026

GROUP 5

- Domain:– Web Dev
- Problem Statement Title:- Real-Time Collaborative Code Editor
- Team Leader Name:- Kaustubh Chauhan
- Group No:- Group 5
- Team Name:- HashHustlers
- Members:- 4

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Real-Time Collaborative Code Editor

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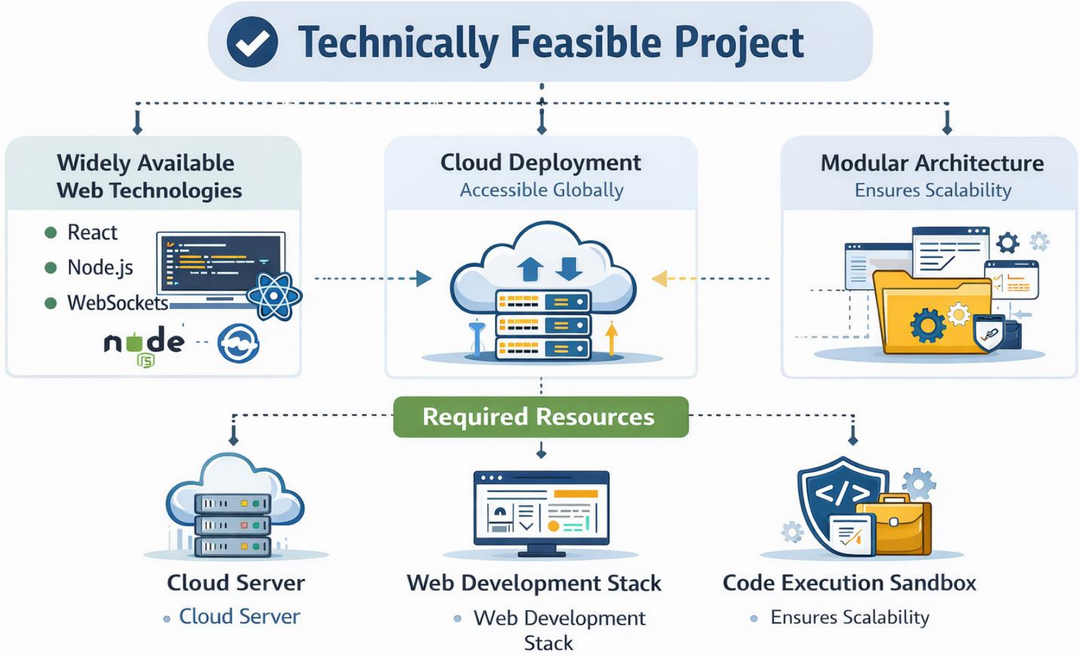
Code Sync is a real-time collaborative coding platform where multiple developers write, edit, and execute code simultaneously. It enables seamless pair programming, remote technical interviews, collaborative debugging, and team-based development.

The platform integrates:

- Conflict-free real-time editing using CRDT (Yjs) over WebSocket
- Shared code execution: JavaScript, Python, C++, Java
- Room-based collaboration with invite links and participant tracking
- Firebase auth, Firestore persistence, and extensible AI architecture

KEY FEATURES

Real-Time Collaborative Editing
•Conflict-free editing with CRDT (Yjs) + collaborative cursor awareness
Live Code Execution
•Run JavaScript, Python, C++, Java — output shared across all collaborators
Room-Based Collaboration
•Create or join rooms via invite link with participant tracking
Firebase Cloud Integration
•Authentication, Firestore persistence, execution history, participant tracking



WOW FACTOR

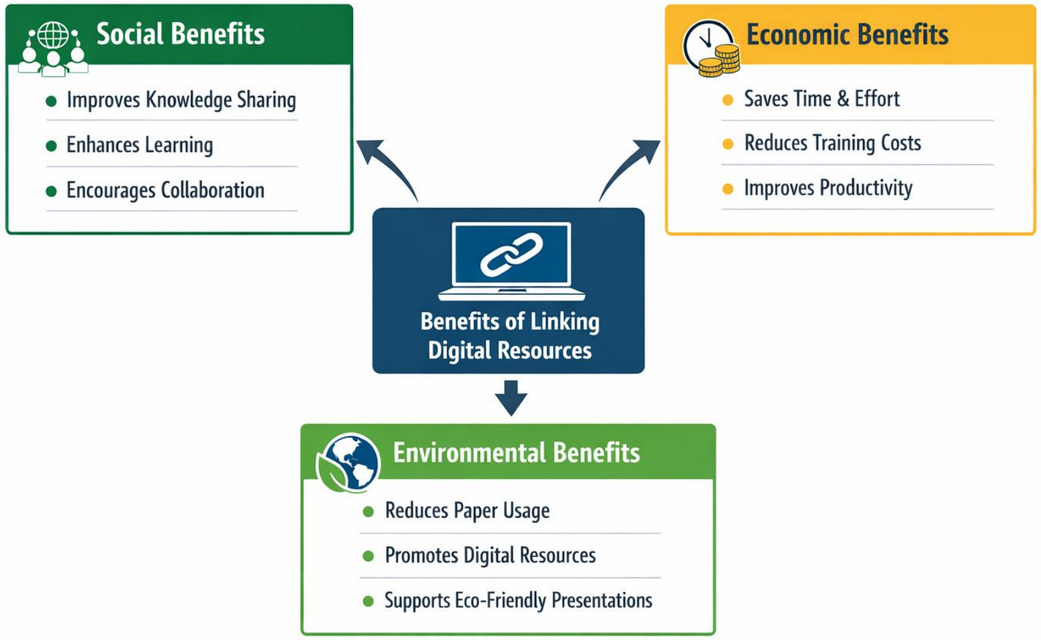
- **What makes Code Sync stand out**

➢ **Zero-Conflict Editing with CRDT**
Yjs CRDT ensures edits from multiple users never conflict — no locking, no waiting just seamless collaboration.

➢ **Shared Live Terminal**
One user runs code and every collaborator instantly sees the output — bridging th gap between writing and execution.

➢ **AI-Ready Architecture**
Built-in hooks for AI code assistant, bug detection, explanation, and test generatio — ready to plug in and ship.

➢ **Monaco Editor — Same as VS Code**
Developers get the full power of a professional editor — syntax highlighting, IntelliSense-style completions — right in the browser.



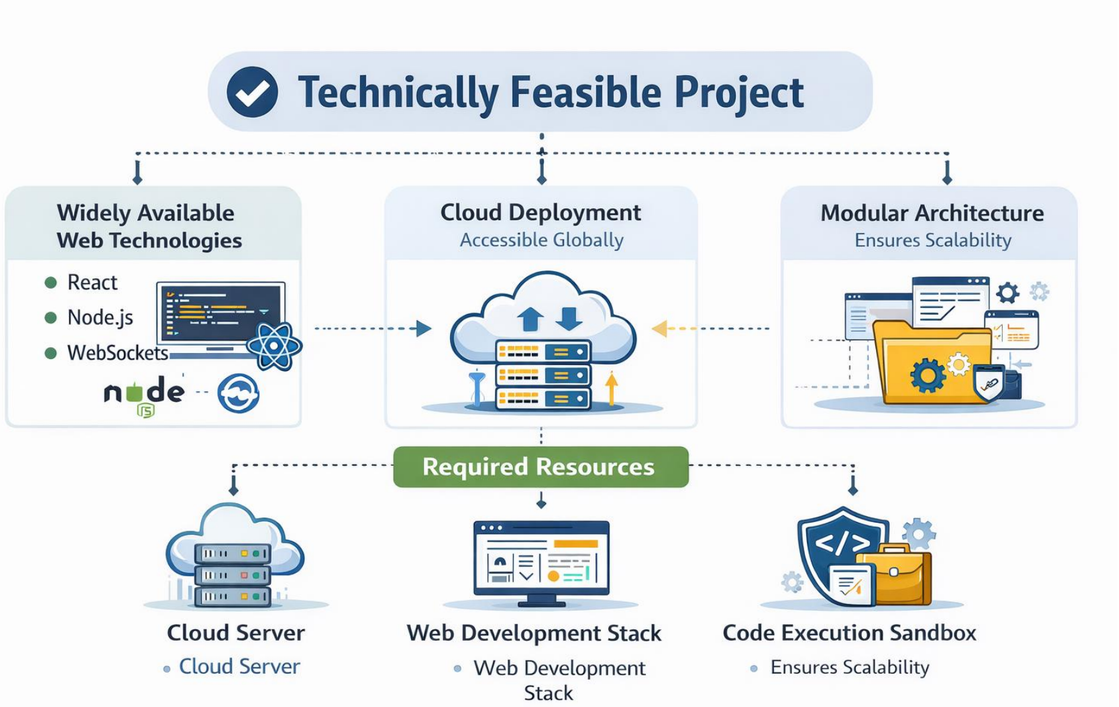
FEASIBILITY AND VIABILITY

The project is technically feasible because:

- Uses widely available web technologies (React, Node.js, WebSockets)
- Cloud deployment makes it accessible globally
- Modular architecture ensures scalability

Required resources:

- Cloud server
- Web development stack
- Code execution sandbox



IMPACT AND BENEFITS

- **Potential impact on the target audience**

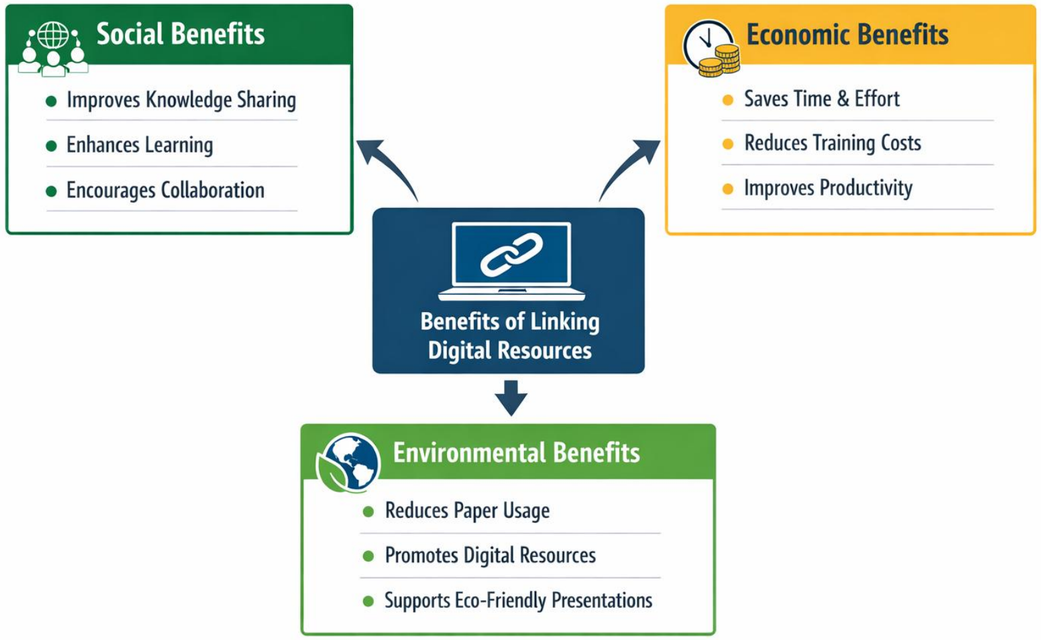
➢ **Improved Remote Collaboration**
Developers from different locations can write and edit code together in real time.

➢ **Better Learning for Students**
Students learning programming can collaborate with teachers or classmates live.

➢ **Faster Problem Solving and Debugging** Multiple developers can analyze and fix coding errors together instantly, which reduces development time.

➢ **Useful for Technical Interviews** Companies can conduct live coding interviews where both interviewer and candidate write and test code on the same platform.

➢ **Supports Open Source Collaboration**
Developers from around the world can work on projects together, making open-source development easier and faster.



RESEARCH AND REFERENCES

Research Papers

- **Real-Time Collaborative Code Editor**

  Author:-DR.P. Manikandaprabhu, S.Shwetha

- **CodeR: Real-time Code Editor Application for Collaborative Programming**

- https://www.researchgate.net/publication/283161500_CodeR_Real-time_Code_Editor_Application_for_Collaborative_Programming

Existing solutions

- **Replit(Multilayer)**
  https://docs.replit.com/replit-workspace/workspace-features/multiplayer

- **CodeSandbox**
  https://codesandbox.io/

- **VS code live share**
  https://visualstudio.microsoft.com/services/live-share/